THE CONORMAL SEQUENCE OF A COMPOSED IMMERSION OF SPACES IS EXACT AND ETALE-LOCALLY SPLIT WHEN THE INNER PIECE IS H_1-REGULAR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10624-10635 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10637-10648, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The schemes-chapter twin of this label (Divisors side) is the cited input, not a restatement of a sibling chapter card; grep census confirms the label exists only spaces-side in the pool chapters.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma transitivity-conormal (the base right-exact sequence); chapter lemmas quasi-regular-immersion and regular-quasi-regular-immersion (the H_1-regular dictionary); Divisors, lemma transitivity-conormal-quasi-regular (the schemes exactness-and-splitting theorem); etale descent of exactness for quasi-coherent modules on spaces.

Source reference. Original lemma 063N: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10648–10648. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09RY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10605–10613. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06BP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10550–10557. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06AE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 591–603. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $Z \to Y \to X$ be immersions of algebraic spaces over S. Assume that $Z \to Y$ is H_1-regular. Then the canonical sequence of Lemma 06AE*

$$0 \to i^*\mathcal{C}_{Y/X} \to \mathcal{C}_{Z/X} \to \mathcal{C}_{Z/Y} \to 0$$

is exact and (étale) locally split.

Proof. Since $\mathcal{C}_{Z/Y}$ is finite locally free (see Lemma 09RY and Lemma 06BP) it suffices to prove that the sequence is exact. It suffices to show that the first map is injective as the sequence is already right exact in general. After étale localization on X this reduces to the case of schemes, see Divisors, Lemma 063N. □